Yang-Jun Joo, Ph.D.

Data Scientist | Quantitative Risk Assessment

Autonomous Vehicle Safety | Statistical & Probabilistic Modeling | Large-Scale Driving Data

Orlando, FL (Open to relocation to Foster City, CA) | **Email** | Phone: +1 (407) 823-2841 | **LinkedIn** | **GitHub** | **Google Scholar**

Professional Summary: Quantitative risk assessment data scientist with 8+ years of relevant research and applied project experience, including 3+ years of post-Ph.D. experience, developing probabilistic safety metrics, uncertainty-aware models, and automated risk-assessment tooling for autonomous-vehicle and transportation systems. Author of 15 published or accepted papers and PI on two competitively funded safety projects, with experience in Python, SQL, large-scale driving data, and human-behavior experiments.

TECHNICAL SKILLS

Programming & Data:	Python, SQL, R, pandas, NumPy, scikit-learn, PyTorch, Git, Linux, ETL and analytics pipelines.
Risk & Statistics:	Probabilistic modeling, Bayesian networks, reliability analysis, surrogate safety measures, regression, treatment-effect analysis, uncertainty quantification, conformal prediction.
Safety & Mobility:	Risk-assessment frameworks, safety-performance metrics, AV safety evaluation, naturalistic driving video, connected-vehicle and trajectory data, traffic-conflict analysis, human factors.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar — **University of Central Florida** — Mar 2024 – Present

- Led FDOT quantitative safety workstreams with cross-functional agency and technical partners, defining and validating safety-performance metrics using high-resolution traffic-controller, connected-vehicle, and naturalistic driving data.
- Built Python and SQL pipelines for multi-terabyte datasets, analyzing high-resolution signal-controller logs from 1,000+ intersections and trajectory data across four freeways to automate and standardize risk-metric generation and reporting.
- Developed uncertainty-aware evaluation methods for autonomous-driving vision-language models using conformal prediction, identifying context-specific calibration failures in safety-critical driving-scene question answering.
- Designed and analyzed a study of 926 participants and 1,325 clips, using regression and treatment-effect analysis to quantify automation effects on passenger risk perception.

Postdoctoral Researcher — **Seoul National University** — Mar 2023 – Feb 2024

- Led research on real-time traffic-safety information integration for the Korean National Police Agency, defining safety-critical scenarios and operational safety criteria.
- Developed system models and evaluation criteria for cooperative traffic-control strategies in mixed-autonomy environments, supporting real-time safety and operational decision-making.

Graduate Research Assistant — **Seoul National University** — Mar 2018 – Feb 2023

- Developed probabilistic and reliability-based risk-assessment models for autonomous-vehicle lane changes, individual-driver crash risk, and proactive collision avoidance using Bayesian networks, surrogate safety measures, and risk-field methods.
- Secured and led a KRW 96.1M Global Ph.D. Fellowship, producing three first-author peer-reviewed publications.

SELECTED PUBLICATIONS

- Joo et al. (2023), "A Generalized Driving Risk Assessment Method for Autonomous Vehicles Using Field Theory," *Analytic Methods in Accident Research*, 37, 100251.
- Joo et al. (2022), "A Data-Driven Bayesian Network for Probabilistic Crash Risk Assessment of Individual Driver with Traffic Violation and Crash Records," *Accident Analysis & Prevention*, 174, 106721.
- Joo et al. (2026), "Context-Aware Conformal Prediction for VLM-Based Driving Scene QA," *Forty-Third International Conference on Machine Learning (ICML) Workshop EMM-QA*.

EDUCATION

Ph.D., Civil & Environmental Engineering (Transportation), Seoul National University, 2023

B.S., Civil & Environmental Engineering, Seoul National University, 2018